

ADWOA ASANTEWAA BREMANG

adwoabremang@gmail.com | +233 26 507 3522 | [linkedin.com/in/asantewaabremang/](https://www.linkedin.com/in/asantewaabremang/) | <https://asantewaa-bremang.github.io/>

EDUCATION

Carnegie Mellon University Africa (CMU-Africa)

MSc. Electrical and Computer Engineering

Kigali, Rwanda

May 2021

Ashesi University

BSc. Electrical Engineering (Magna Cum Laude)

Eastern Region, Ghana

June 2019

AWARDS

Full Scholarship Recipient, Mastercard Foundation – Carnegie Mellon University Africa

Aug 2019

PUBLICATIONS

- **Bremang, A. A.**, Owusu, D. A., Quagraine, V. K., & Annor-Adjaye, L. (2026). *Creating Task-Specific Speech Recognition Datasets from Scratch for Low-Resource Languages: Assessing the Impact of Token Sequence Overlap*. In Proceedings of LREC 2026 (Accepted).
- **KEITA, M. K., Zampieri, M., Homan, C. M., Bremang, A. A., Owusu, D. A., & Le, H.** (2026). *Grammatical Error Correction for Low-Resource Languages: The Case of Zarma*. **Accepted to** The Second Workshop on Language Models for Low-Resource Languages (LoResLM 2026), Rabat, Morocco.

PROFESSIONAL EXPERIENCE

Ashesi University

Accra, Ghana

Research Support Staff – Computer Science Department

January 2024 – Present

- Rebuilt Fairseq's unsupervised Wav2vec (Wav2vec-U) pipeline with streamlined installation scripts, enabling unsupervised speech recognition experimentation for any language — applied to Twi as a low-resource case study.
- Developing TWI and accented English speech-to-text models to assist students with their final projects.
- Contributed to creating a grammar error correction model for Zarma, a low-resource language.
- Supervised a student capstone project, providing guidance, critical feedback, and support for research in human activity recognition using cameras. The student group won the award for Best Group Project.
- Co-instructed a deep learning class of 30+ students on foundational concepts in Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs).

Data Scientist – Ashesi E-learning

April 2022 – Feb 2024

- Built an NLP speech-to-text model that surpassed Google's speech-to-text on an African-accented dataset by approximately 30% word error rate.
- Led a team of over 100 contributors to develop a 40-hour dataset of African-accented English.

Carnegie Mellon University Africa

Kigali, Rwanda

Teaching Assistant

Sep 2020 – Aug 2021

Course - Orientation Programming Session

- Trained over 100 students in Java and Linux programming, providing personalized support during office hours to address individual weaknesses, significantly boosting student confidence in programming.

PROJECT & RESEARCH EXPERIENCE

Solid Rock Foundation, Educational Project

Accra, Ghana

Software Engineer – Educational institution data management

October 2023 - present

- Developed a Python-based data analysis tool that generated terminal reports for approximately 100 elementary and high school students for over a year, enabling teachers to prioritize time on student support.

Datagon Studio, API development Project

Accra, Ghana

Consultant – Computational API

March 2024 - Present

- Developed a computational API for assessing educational institutions for an educational consulting firm.
- Translated Excel-based computational processes into a Flask API using Python and Google Cloud Platform.

LEADERSHIP EXPERIENCE

Deep learning Indaba, IndabaX Ghana Berekuso, Ghana
Tutorial facilitator June 2025, June 2026

- Co-facilitated a tutorial on “Foundational NLP: Pretraining and Key Concepts and Introduction to Python for Data” at IndabaX Ghana 2025-2026 to a crowd of 15+ participants

EnhanceMind AI Conference 2025, AI Conference Dar es Salam, Tanzania
Keynote Speaker September 2025

- Delivered a virtual talk on *Applications of NLP*, highlighting the evolution of NLP and Africa’s role in its future development.

SKILLS & INTERESTS

Technical Skills: Deep Learning, Programming, IoT, Research

Tools: *Programming:* C++, Java, Linux, NumPy, Python, Google Cloud Platform, PyTorch

Interests: Community Service, Machine Learning